Functional Imaging of Pelvic Floor Muscles

Ultrasound Scanning of Asymptomatic Subjects

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Methodology

1. Purpose: To gain understanding of the mechanisms of pelvic floor action in terms of displacement velocity and acceleration
2. Perineal Ultrasound Scanning (5.5 Mhz Elliptical Probe)
3. Number of subjects: 23
4. Age Range: 31.1 ± 13.6 years
5. Body Mass: 22.4 ± 1.9 kg
6. Segment: Urethra Displacement, and Anorectal Junction Motility

Data Collection Analysis

- Video Recordings of Imaging Signals being recorded on a PC for offline analysis by a trained physician
- Analysis of the data that was performed offline
- Velocity was analyzed by differentiating displacement/time
- Acceleration was computed by re-differentiating velocity/time

Effect on bladder/urethra and effect on anorectal junction(ARJ)

PFN(Active), Valsalva(Passive), Cough(Transient)

Voluntary Activation (PFM Precontraction / “The Knack”)

- **Bladder/Urethra**: Increases urethral closure pressure; elevates bladder neck; prevents urine leakage during stress
- **Anorectal Junction (ARJ)**: Cranial lift and stabilization of pelvic floor structures

Passive Activation (Valsalva)

- **Bladder/Urethra**: Downward descent of bladder neck and urethra; increased risk of leakage/funneling
- **Anorectal Junction (ARJ)**: Caudal (downward) displacement and organ descent

Transient Activation (Cough)

- **Bladder/Urethra**: Rapid rise in intra-abdominal pressure → urine loss without protection; “The Knack” reduces leakage by 73–98%
- **Anorectal Junction (ARJ)**: Brief reflexive pelvic floor response; enhanced by pre-contraction

Key Finding (Miller et al., 1998)

- Timed PFM contraction (“The Knack”) before/during cough significantly reduces cough-related urine loss in women with mild SUI within 1 week

Evaluation through perineal ultrasound

Pelvic Floor Maneuver(Arrow for movement)







Evaluation through perineal ultrasound

Valsalva(Arrow for movement)







Evaluation through perineal ultrasound

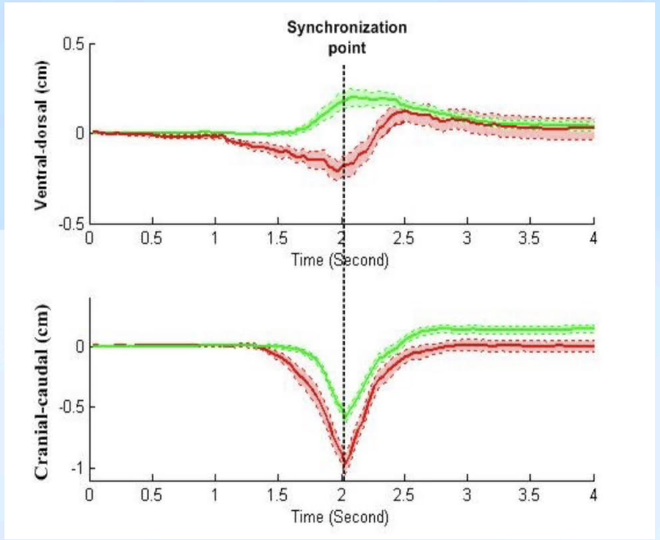
Cough(Arrow for movement)



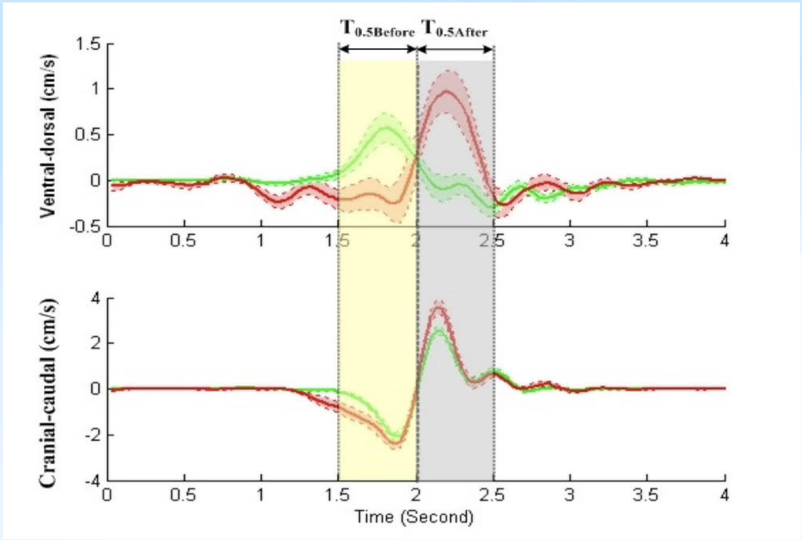




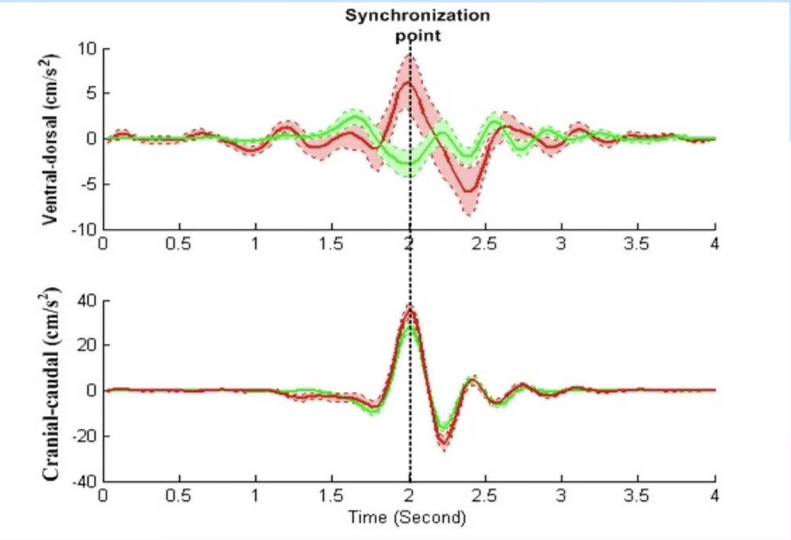
Evaluation for Cough Induced Stimulation of Physical Quantities



Displacement



Velocity



Acceleration

Function during Voluntary, Passive, and Transient activation

Contraction (Active) Valsalva (Passive) and Cough (Transient)

Voluntary Activation (Contraction)

- Conscious contraction
- Elevates pelvic organs, narrows elevator hiatus, increases urethral and anal closure pressure
- Key for strengthening and continence training

Passive Activation (Valsalva)

- Increased intra-abdominal pressure with downward push
- Tests pelvic floor support and descent
- Used to assess prolapse and organ mobility

Transient Activation (Cough)

- Rapid, reflexive co-contraction of pelvic floor muscles
- Counters sudden rise in intra-abdominal pressure
- Critical for dynamic continence during daily activities

Interpretation of Imaging

- Voluntary Pelvic Floor Muscle Contraction results in End Compression of the Urethra
- The results show that this action provides greater support for the urethra
- There are no significant CAUDAL displacement during a cough

Take Home Message

- Perineal ultrasound scanning provides a non-invasive imaging of dynamic pelvic floor function
- Image analysis documents numerical parameters of passive and dynamic activation of the pelvic floor

References